

MAD & PWA Lab

Journal

Experiment No.	05
Experiment Title.	To apply navigation, routing and gestures in Flutter App
Roll No.	02
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Class	D15A/D15B
Subject	MAD & PWA Lab
Lab Outcome	LO2: Design and Develop interactive Flutter App by using widgets, layouts, gestures and animation
Grade:	

Experiment No. 5

Aim:-

To apply navigation, routing and gestures in Flutter App

Theory:-

Navigation in Flutter allows users to move between different screens (or pages) in the app. Flutter uses the Navigator widget to handle navigation between routes (screens).

Types of Navigation

- Push Navigation (Forward Navigation) → Moves to a new screen.
- Pop Navigation (Backward Navigation) → Moves back to the previous screen.
- PushReplacement → Replaces the current screen with a new one.
- PushAndRemoveUntil → Moves to a new screen and removes previous screens from the stack.

Types of Routing

1. Direct Route Navigation (MaterialPageRoute)-Used for simple page-to-page navigation. 2. Named Routes (Predefined Routes in main.dart)-Defined in the MaterialApp widget and used throughout the app.

Flutter uses the GestureDetector widget to detect user interactions like taps, swipes, pinches, and long presses. This is essential for making an app interactive.

Common Gestures & Their Uses:

- Tap → Detects simple taps on a widget.
- Double Tap → Recognizes double-clicking.
- Long Press → Triggers an action when the user presses and holds.
- Swipe (Drag) → Detects horizontal or vertical dragging.
- Pinch (Zoom In/Out) → Detects two-finger pinch for zooming.

Code:-

```
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';
import 'package:flutter_svg/flutter_svg.dart';
import 'package:foodie/constants/app_constants.dart';
import 'package:foodie/router/scaffold_with_bottom_nav_controller.dart';
import 'package:go_router/go_router.dart';
```

```
import '../constants/string_constants.dart';
import '../features/recipes/presentation/recipe_controller.dart';
import '../providers/providers.dart';
import '../theme/theme.dart';
import 'app_route.dart';
```

```
class ScaffoldWithBottomNavBar extends ConsumerStatefulWidget {
  const ScaffoldWithBottomNavBar({
    required this.child,
    super.key,
  });
  final Widget child;
```

```
  @override
  ConsumerState<ScaffoldWithBottomNavBar> createState() =>
    _ScaffoldWithBottomNavBarState();
}
```

```
class _ScaffoldWithBottomNavBarState
  extends ConsumerState<ScaffoldWithBottomNavBar> {
  void _tap(int index) {
    ref.read(scaffoldBottomNavControllerProvider.notifier).setPosition(index);
    ref.invalidate(ingredientMultiplierProvider);
  }
}
```

```

ref.invalidate(instructionsCounterProvider);
if (index == 0) {
  context.goNamed(AppRoute.home.name);
  ref.read(recipeControllerProvider.notifier).ref.invalidateSelf();
} else if (index == 1) {
  context.goNamed(AppRoute.challenges.name);
} else if (index == 2) {
  context.goNamed(AppRoute.saved.name);
} else if (index == 3) {
  context.goNamed(AppRoute.friends.name);
} else {
  context.goNamed(AppRoute.profile.name);
}
}
}

```

@override

```

Widget build(BuildContext context) {
  int position = ref.watch(scaffoldBottomNavControllerProvider);
  return Scaffold(
    body: widget.child,
    bottomNavigationBar: BottomNavigationBar(
      currentIndex: position,
      onTap: (index) => _tap(index),
      items: [
        _buildBottomNavBarItemWidget(Assets.icons.home, StringConstants.home),
        _buildBottomNavBarItemWidget(
          Assets.icons.challenges, StringConstants.challenges),
        _buildBottomNavBarItemWidget(
          Assets.icons.saved, StringConstants.saved),
        _buildBottomNavBarItemWidget(

```

```

        Assets.icons.friends, StringConstants.friends),
    _buildBottomNavBarItemWidget(
        Assets.icons.profile, StringConstants.profile),
    ],
),
);
}

```

```

BottomNavigationBarItem _buildBottomNavBarItemWidget(
    String iconName, String label) {
    return BottomNavigationBarItem(
        icon: SvgPicture.asset(
            iconName,
            height: AppConstants.bottomNavBarIconHeight,
        ),
        activeIcon: SvgPicture.asset(
            iconName,
            height: AppConstants.bottomNavBarIconHeight,
            theme: const SvgTheme(
                currentColor: ThemeColors.primary,
            ),
            colorFilter:
                const ColorFilter.mode(ThemeColors.primary, BlendMode.srcATop),
        ),
        label: label);
    }
}

```

Output:-

Hello, Mateo

Prepare delicious meals
and win prizes!



Search recipes

Search for: chicken



Sweet & Sour Chicken

10 points



Asparagus-Stuffed Chicken Breast

10 points



Low-Carb Mediterranean Chicken And Tomato By

10 points 30 mins



Easy Butter Chicken

10 points 50 mins

← Add friend

Search the community



Marija



Laura



Inga



test



Home



Challenges



Saved



Friends



Profile

Saved recipes



Jacket Potato: The Old Fashioned

10 points



Vegan Butternut Squash Al Pastor Tacos

10 points



How To Make A Fresh Seacuterie Board

10 points



Home



Challenges



Saved



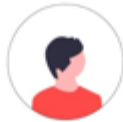
Friends



Profile

14:16

35



Mateo

Points 0

Level 1



Medals



Rewards



Future chef

Make one meal



Iron chef

Complete 3 challenges



Prep master

Save 2 recipes



A silver pot

Reach level 2



Golden pasta

Reach level 3



Home



Challenges



Saved



Friends



Profile

Culinary missions

August 28 - September 3



Beginner

Make a dish with eggs

0 1

10 points



Veggie lover

Use tomato or salad in 3 dishes

0 3

15 points



Chicken master

Use the chicken in 3 dishes

0 3

15 points



Home



Challenges



Saved



Friends



Profile